Polite Bursting: Feedback Admission Control under Age-Limited Sensing on Shared Clusters

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Abstract*—Research and AI teams burst GPU/ML batch workloads onto shared, multi-tenant clusters where they are *guests*: bound by fair-use policy (pod caps, utilization floors) and forced to sense a neighbour-driven, unobservable capacity through a *stale* probe. Naive submission violates policy; static rate-limiting wastes throughput; and adaptive control sometimes helps and sometimes does not. We explain *when* by casting polite bursting as chance-constrained admission under age-of-information: the controller acts on a capacity estimate of age D. Our central result is an exact *feedback-usefulness limit*: for a symmetric two-state capacity, the goodput advantage of any causal admission controller over the best static budget equals *half the autocorrelation of available capacity at the sensing lag*, $\Delta(D) = \frac{1}{2}r(D)$—a value that stays *achievable* for any symmetric stationary capacity (via a memoryless threshold), with the general converse governed by the capacity's β-mixing rate. It yields a sharp impossibility (no controller beats static once D exceeds the neighbour's coherence time τ_c), a matching achievability, and a *regime-adaptive* controller that estimates τ_c online and switches between closed- and open-loop admission to track the limit uniformly. We validate the law in simulation—the optimal policy sits on $\frac{1}{2}r(D)$ to within 0.013, practical AIMD degrades below static exactly when $\tau_c \lesssim D$, and the regime-adaptive controller never falls below static (a live GPU run confirms its online $r(D)$ estimate lands on the curve and the switch's politeness dividend grows as feedback fails)—and on a workstation testbed under *real, manufactured GPU contention* (with the full submission path verified end-to-end on NRP Nautilus): under structured contention feedback admission is the polite-efficient knee, $1.41\times$ static's goodput at $4\times$ lower over-admission *than naive* while sparing the competing tenant a greedy policy harms, and up to $3.2\times$ under a self-imposed budget. The pre-registered memoryless regime where it breaks even is exactly the $r(D) \to 0$ prediction—the regime the adaptive controller is built to detect and sidestep. The system and a reproducible artifact are available to reviewers.

I. Introduction

A growing fraction of machine-learning work is *bursty*: long stretches of local development punctuated by short, intense bursts of batch jobs (sweeps, fine-tunes, evaluations) that briefly need far more GPUs than a team owns. Shared, opportunistic clusters such as the National Research Platform (NRP) [1] make that capacity available, but on strict *guest* terms: fair-use policy caps concurrent pods, demands sustained utilization of what you request, bans sleep-only or idle workloads, and reclaims resources on completion. The guest is also *networked* as an outsider—often reachable only through a single forwarded port or a mesh VPN, with no inbound cluster access.

In this setting the two obvious strategies both fail. *Naive* submission fires all work at once: it grabs more than its fair share, leaves pods pending or under-utilized, and trips the very thresholds that get a namespace banned. *Static* rate-limiting is safe but leaves the budget idle whenever capacity is available, wasting the guest's own throughput. Neither adapts.

We argue that being a good guest is a *control* problem whose answer is set by *how stale the guest's view of the cluster is*, and we build that thesis into PoliteBurst, a container-native bursting fabric. Our contributions:

- An **age-of-information formulation** of polite bursting (§III) and a **feedback-usefulness limit** (§IV): with a capacity estimate of age D, the goodput advantage of *any* causal admission controller over the best static budget is $\Delta(D) = \frac{1}{2}r(D)$, half the autocorrelation of available capacity at the sensing lag—exactly so for a symmetric two-state capacity (Thm. 1), and achievably in general, with the converse governed by β-mixing (App. A)—an impossibility once D exceeds the neighbour's coherence time τ_c (Cor. 1), lifted to any pod cap by a layer-cake reduction (Prop. 2).
- A **regime-adaptive controller** (§V) that estimates τ_c online from the probe stream and switches between closed- and open-loop admission, provably tracking the limit uniformly in τ_c (Thm. 2) with a deterministic hard-cap safety guarantee (Prop. 1).
- A **realization** (§VI) in which the controller runs in-cluster and senses/actuates across a single-port WAN leaf link that fixes the sensing age D; the fabric's own design and characterization are the subject of a companion systems paper [2] and are out of scope here.
- An **evaluation** (§VIII) that validates the law in simulation (optimal policy on $\frac{1}{2}r(D)$ within 0.013; adaptive control never below static) and, on a workstation testbed under *real GPU contention* (full submission path verified end-to-end on NRP Nautilus), shows feedback admission is the polite-efficient knee that spares the competing tenant—while the pre-registered memoryless regime where it breaks even is exactly the $r(D) \to 0$ prediction the adaptive controller detects and sidesteps.

II. Background and Motivation

Host policy. The reference host (NRP Nautilus [1]) publishes a fair-use policy that our design treats as ground truth: at most $K{=}4$ concurrent pods per user; per running pod, utilization relative to *request* must stay in published bands (GPU $\geq 40\%$, CPU 20–200%, RAM 20–150%); jobs must do real work (sleep-only jobs are banned) and release resources on completion; idle storage volumes are reclaimed. Admins

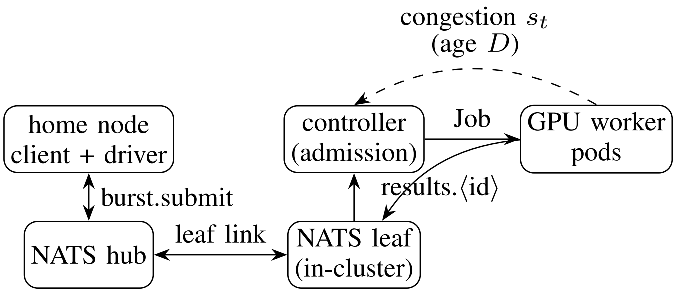


Fig. 1. PoliteBurst architecture. The home client submits over a federated NATS leaf link; the in-cluster controller admits jobs (pacing concurrency to the politeness budget) and launches GPU worker pods, whose results route back through the leaf. The dashed arc is the stale feedback the controller acts on: an age-D congestion signal s_t whose delay bounds the achievable advantage (Thm. 1).

monitor utilization and ban namespaces that hold under-utilized requests—so politeness is not optional.

Network reality. The guest reaches the cluster over a single PAT (port address translation)-forwarded port or a mesh VPN, with no inbound access. A control plane must therefore live behind that boundary and communicate over a messaging fabric that traverses it—motivating the NATS leaf-node design of §VI.

Why existing tools are not enough. Workflow and function-execution frameworks—HTCondor [3], funcX/Globus Compute [4], Parsl [5], Ray [6], Dask [7]—scale workers to demand but carry no notion of a host *politeness budget*; left to default scaling they over-provision under contention. We instead make self-restraint a first-class, controlled objective.

III. System Model: Polite Bursting under Age-Limited Sensing

A *guest* bursts batch GPU pods onto a shared cluster and is bound by a fair-use policy: at most K concurrent pods, and each admitted pod must sustain a utilization floor (else it is a poor citizen, risking throttling or a ban). Time is slotted into control epochs $t \in \mathbb{N}$. Let $a_t \in \{0, \dots, C\}$ be the capacity the cluster can *productively* give the guest at epoch t; it is set by the other tenants—exogenous, stochastic, and **not directly observable**. The guest chooses a window $w_t \in \{0, \dots, K\}$. Useful goodput is $G_t = \min(w_t, a_t)$; the surplus $(w_t - a_t)^+$ is *over-admitted*: those pods sit below the utilization floor and mark the guest impolite. With $\rho \triangleq \limsup_T \frac{1}{T}\sum_{t<T} \mathbf{1}[w_t > a_t]$ the long-run over-admission fraction, the guest solves

$$\max_{\{w_t\}} \ \mathbb{E}[\min(w_t, a_t)] \quad \text{s.t.} \quad \underbrace{w_t \le K \ \forall t}_{\text{hard cap}}, \qquad \underbrace{\rho \le \epsilon}_{\text{policy}}. \tag{1}$$

The constraint is *two-sided*—stay below the cap yet above the floor—which already separates it from one-sided (loss-only) congestion control.

a) Age-limited sensing.: The guest does not observe a_t; it *probes* cluster state, and the probe is stale. We model this as an *age of information* D: at epoch t the controller's freshest reading of capacity has age D, i.e. admissible policies are causal functions of $\{a_{s-D}\}_{s\le t}$. D is a physical, *measured* quantity (Sec. VIII: $D \approx 2.5\,\mathrm{s}$, dominated by CUDA cold-start, not the $\sim 27\,\mathrm{ms}$ poll). Age-limited sensing is the crux: a controller can only act on a D-old picture of a world the neighbours are still changing.

b) The contended unit.: Politeness is decided at the margin. Fix attention on the single contended resource unit whose availability fluctuates with the neighbour, and let $x_t \in \{0, 1\}$ indicate that it is *free* ($x_t = 1$) or *taken* ($x_t = 0$); admitting it when taken is the over-admission the policy punishes. We take x_t stationary with $\Pr[x_t = 1] = \frac{1}{2}$ (a fully contended unit; the asymmetric case only rescales the constants below). The guest's marginal decision $m_t \in \{0, 1\}$ is a (possibly randomized) causal function of the age-D observation $\hat{x}_t \triangleq x_{t-D}$. Marginal goodput and over-admission are

$$g \triangleq \Pr[m_t = 1,\, x_t = 1], \qquad \rho \triangleq \Pr[m_t = 1,\, x_t = 0]. \tag{2}$$

Everything below concerns the achievable (ρ, g) region under age D. (The stationary $\rho = \Pr[m_t{=}1, x_t{=}0]$ of (2) is the ergodic limit of the epoch fraction in (1); we reuse ρ for both, and write $\bar{\rho}$ for the surplus *amount* in Prop. 2.)

Proposition 1 (Hard-cap safety)**.** *Any window update that clamps additive increase at K and never increases w_t on a congestion signal keeps $w_t \le K$ for all t deterministically; the pod-cap is never violated.*

Proof. Induction on t: increase is clamped to K, decrease cannot exceed the current value. □

IV. The Feedback-Usefulness Limit

We now characterize *exactly* how much closed-loop admission can gain over open-loop (static) admission when the only signal is a D-old observation. The answer is a single, interpretable quantity: the autocorrelation of capacity at the sensing lag.

Lemma 1 (Predictive disagreement)**.** *Let $q(D) \triangleq \Pr[x_t \ne x_{t-D}]$ be the lag-D disagreement probability and $r(D) \triangleq \mathbb{E}[(2x_t - 1)(2x_{t-D} - 1)]$ the lag-D autocorrelation of the centered availability. Then $r(D) = 1 - 2q(D) \in [-1, 1]$, and by stationarity and symmetry the joint law of $(\hat{x}_t, x_t)$ is $\Pr[\hat{x}_t{=}i, x_t{=}j] = \frac{1}{4}\big(1 + (2i{-}1)(2j{-}1)\, r(D)\big)$. For a symmetric two-state Markov capacity with per-epoch flip probability $p \in (0, \frac{1}{2})$, $r(D) = (1 - 2p)^D = e^{-D/\tau_c}$ with coherence time $\tau_c \triangleq -1/\ln(1 - 2p) \approx 1/(2p)$.*

Proof. $r = \mathbb{E}[(2x_t - 1)(2x_{t-D} - 1)] = \Pr[\text{agree}] - \Pr[\text{disagree}] = 1 - 2q$. The joint follows from the two symmetric marginals and the single correlation. For the Markov chain, writing $\lambda_2 \triangleq 1 - 2p$ for the second eigenvalue, the D-step transition is $P^D = \frac{1}{2}\begin{pmatrix} 1+\lambda_2^D & 1-\lambda_2^D \\ 1-\lambda_2^D & 1+\lambda_2^D \end{pmatrix}$, so $\Pr[\text{disagree}] = \frac{1}{2}(1 - \lambda_2^D)$ and $r = \lambda_2^D = (1 - 2p)^D$ (the two-state chain's autocorrelation and its total-variation distance to stationarity $\frac{1}{2}(1 - 2p)^D$ are standard mixing facts [8]). □

Theorem 1 (Achievable region and the feedback advantage)**.** *For the symmetric two-state Markov capacity of Lemma 1, under age-D sensing the achievable (ρ, g) region of* (2) *is the quadrilateral with vertices* $\{(0,0), (\frac{q}{2}, \frac{1-q}{2}), (\frac{1}{2}, \frac{1}{2}), (\frac{1-q}{2}, \frac{q}{2})\}$ *($q = q(D)$), whose efficient (upper) boundary is the polyline* $(0,0) \to (\frac{q}{2}, \frac{1-q}{2}) \to (\frac{1}{2}, \frac{1}{2})$*. The best* static *(observation-oblivious) policy is confined to the diagonal $g = \rho$. Consequently the goodput gain of feedback over static, maximized over matched over-admission levels, is*

$$\boxed{\Delta(D) \;=\; \tfrac{1}{2}\, r(D) \;=\; \tfrac{1}{2}\big(1 - 2q(D)\big)} \tag{3}$$

attained at the kink $\rho = q/2$ by the Bayes threshold policy $m_t = \hat{x}_t$ (at a lower level $\rho < q/2$ the gain is the smaller $\rho(1-2q)/q$); i.e. the peak value of closed-loop admission is exactly half the autocorrelation of available capacity at the sensing lag *(Fig. 2a). The optimality of such memoryless threshold policies for symmetric two-state Markov observation is itself classical in the opportunistic-access setting [9]; what is new here is the exact value $\frac{1}{2}r(D)$ of the resulting feedback advantage over the best static budget.*

Proof. For the symmetric two-state Markov chain the freshest sample $\hat{x}_t = x_{t-D}$ is a sufficient statistic for x_t given the entire delayed history $\{x_{s-D}\}_{s \le t}$ (the Markov property makes older samples conditionally uninformative), so an optimal causal age-D policy depends on the history only through $\hat{x}_t$; we may therefore restrict without loss to policies parameterized by a pair (θ_1, θ_0), $\theta_i = \Pr[m_t{=}1 \mid \hat{x}_t{=}i]$. Using Lemma 1, $g = \frac{1}{2}\big(\theta_1(1-q) + \theta_0 q\big)$ and $\rho = \frac{1}{2}\big(\theta_1 q + \theta_0(1-q)\big)$. This linear map sends the square $[0,1]^2$ to the quadrilateral with vertices at $(\theta_1, \theta_0) \in \{(0,0), (1,0), (1,1), (0,1)\}$, i.e. $(0,0), (\frac{q}{2}, \frac{1-q}{2}), (\frac{1}{2}, \frac{1}{2}), (\frac{1-q}{2}, \frac{q}{2})$; its upper boundary is the stated hull. Maximizing $g - \lambda\rho$ turns on θ_1 before θ_0 because $q < \frac{1}{2}$ makes the free-observation state strictly more goodput-efficient, so the max-goodput point for $\rho \le q/2$ is the kink $(\frac{q}{2}, \frac{1-q}{2})$, achieved by $m_t = \hat{x}_t$. A static policy ignores $\hat{x}_t$ ($\theta_1 = \theta_0 = \theta$), giving $g = \rho = \theta/2$: the diagonal. The vertical gap at $\rho = q/2$ is $\frac{1-q}{2} - \frac{q}{2} = \frac{1-2q}{2} = \frac{1}{2}r(D)$. ☐

Corollary 1 (Impossibility as sensing ages)**.** $\Delta(D) \to 0$ *as $r(D) \to 0$. For the symmetric two-state (Markov) model $\Delta(D) = \frac{1}{2}e^{-D/\tau_c}$, so no causal controller with sensing age D can outperform the static budget once $D \gg \tau_c$: here closed-loop admission helps only to the extent that the guest can sense the neighbour faster than the neighbour changes. In this model low autocorrelation and unpredictability coincide, but in general they need not—a decorrelated process ($r(D){=}0$) can still be exactly predictable from its history (e.g. a random-phase deterministic square wave)—so for an* arbitrary *capacity process the vanishing of the advantage is governed by the β-mixing coefficient, not by $r(D)$. At $D = 0$, $\Delta = \frac{1}{2}$ (capacity is perfectly trackable); the advantage halves every $\tau_c \ln 2$ epochs of added age (mapped in Fig. 2b). App. A makes the general statement precise, extending the impossibility to arbitrary (non-binary, non-Markov) capacity processes with the decay rate set by $\beta(D)$.*

a) Provenance and scope.: The ingredients of (3) are classical, and we claim none of them: the value of information is nonnegative and convex in the prior [10], it is bounded by the payoff oscillation times the total-variation movement of beliefs [11], that movement at sensing age D is precisely the β-mixing coefficient [12], and for the symmetric two-state chain the mixing facts and memoryless-threshold optimality are standard [8], [9]. Our contribution is their assembly under *age-limited, exogenous* sensing—the exact identity $\Delta(D) = \frac{1}{2}r(D)$ with threshold achievability, its β-mixing converse (App. A), and its confirmation on real cluster contention (§VIII). The exogenous structure is essential: unlike the driven-plant setting of networked control, the admission action here does not affect capacity, which is exactly what collapses the value of feedback onto a single predictability coefficient.

Corollary 2 (The square/Poisson experiments are two points on (3))**.** *The contention study of Sec. VIII instantiates exactly this two-state $C{=}2$ model.* Structured (square) *load has $\tau_c \gg D$, so $r(D) \approx 1$ and feedback is near-maximally useful (low ρ, large goodput gain);* memoryless (Poisson) *load drives $\tau_c \lesssim D$, so $r(D) \approx 0$ and feedback collapses onto static—the pre-registered null. The single curve* (3) *predicts both regimes and the crossover between them.*

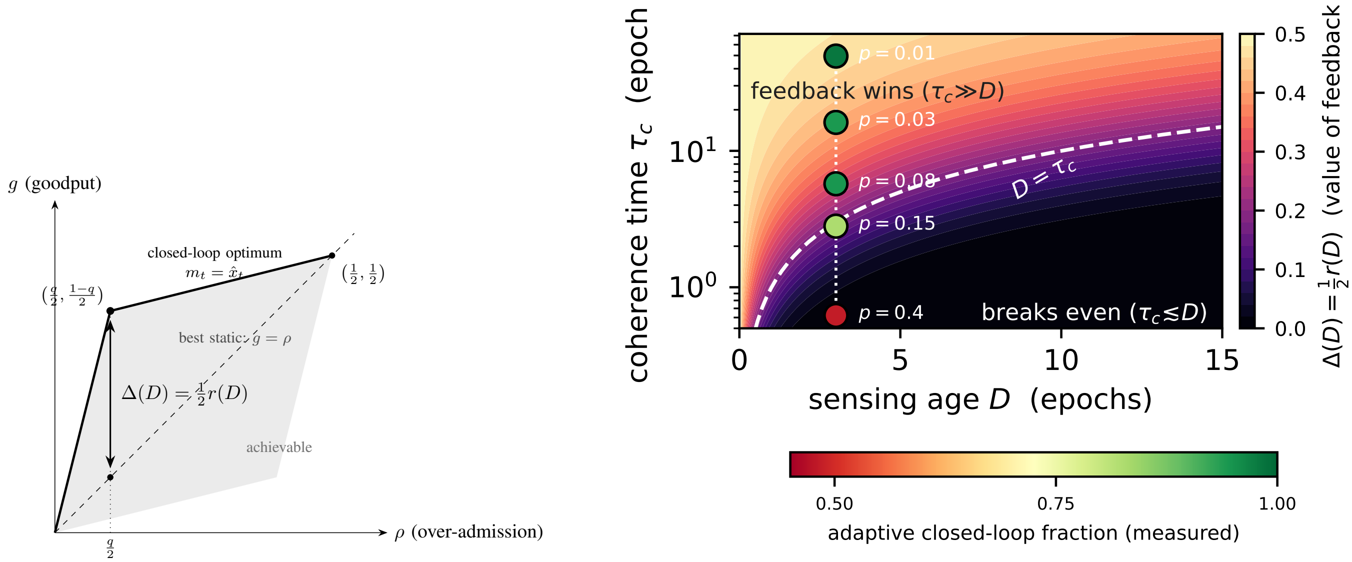
The binary reduction is not a special case but the atom of the general problem.

Proposition 2 (Layer-cake reduction for $C > 2$)**.** *Let $a_t \in \{0, \ldots, C\}$ and score over-admission by the surplus amount $\bar{\rho} \triangleq \mathbb{E}[(w_t - a_t)^+]$. With $x_t^u \triangleq \mathbf{1}[a_t \ge u]$ and $\pi_u \triangleq \Pr[x_t^u = 1]$, both objectives decompose additively over units,* $\min(w_t, a_t) = \sum_{u \ge 1} \mathbf{1}[w_t \ge u]\, x_t^u$ *and* $(w_t - a_t)^+ = \sum_{u \ge 1} \mathbf{1}[w_t \ge u](1 - x_t^u)$*. The age-$D$ optimum is the delayed-capacity window $w_t = a_{t-D}$, and its total goodput advantage over the best comonotone static budget at matched surplus is*

$$\Delta(D) = \sum_{u=1}^{K} \pi_u\, r_u(D), \qquad r_u(D) = \operatorname{corr}\big(x_t^u, x_{t-D}^u\big), \tag{4}$$

which for a single symmetric contended unit ($\pi_u = \frac{1}{2}$) recovers $\Delta = \frac{1}{2}r(D)$.

Proof. The two identities are the layer-cake (Fubini) expansions of $\min$ and $(\cdot)^+$. Setting $w_t = a_{t-D}$ makes $\mathbf{1}[w_t \ge u] = x_{t-D}^u = \hat{x}_t^u$, i.e. the Thm. 1 threshold policy applied independently to each layer; the layers' nesting $x_t^1 \ge x_t^2 \ge \cdots$ is inherited by the observations, so the per-unit thresholds compose into a valid scalar window. For an asymmetric layer with $\Pr[x^u{=}1] = \pi_u$, the same LP as Thm. 1 gives per-unit advantage $\operatorname{Cov}(x_t^u, x_{t-D}^u)/(1 - \pi_u) = \pi_u r_u(D)$ over the static policy matching its marginal; additivity of goodput and surplus sums these to (4). ☐



(a) the feedback advantage is geometry

(b) the impossibility as a map

Fig. 2. **(a)** Under age-D sensing the achievable (ρ, g) pairs form the shaded hull; static policies lie on the diagonal $g = \rho$, and the closed-loop optimum $(\frac{q}{2}, \frac{1-q}{2})$ (memoryless threshold $m_t = \hat{x}_t$) sits a vertical $\Delta(D) = \frac{1}{2}r(D)$ above it (Thm. 1). **(b)** The same value $\frac{1}{2}e^{-D/\tau_c}$ over the sensing-age/coherence-time plane (log τ_c); points are the measured E9 Markov contention sweep (Cor. 1), each shaded by the adaptive controller's closed-loop fraction, opening onto static as D crosses τ_c.

At $C{=}2$ the contended layer alone fluctuates ($\pi_1{=}1, r_1{=}0$; $\pi_2{=}\frac{1}{2}$), so (4) collapses to $\frac{1}{2}r_2(D)$: the binary law is exactly the $C{=}2$ instance, and the experiments measure it directly.

V. Achievability: A Regime-Adaptive Controller

Theorem 1 exposes a gap that a *fixed* controller cannot close: additive-increase/multiplicative-decrease (AIMD) tracks a slowly varying neighbour and approaches the kink $(\frac{q}{2}, \frac{1-q}{2})$ when $\tau_c \gg D$, but *over-admits* when $\tau_c \lesssim D$ (it keeps probing a signal that is pure noise); a static budget is safe everywhere but forfeits $\Delta(D)$ when $\tau_c \gg D$. The optimal policy must know which regime it is in. Since $r(D)$ is itself *measurable from the probe stream*, the guest can estimate it online and switch.

> **Regime-adaptive admission.** Maintain the empirical lag-D correlation $\hat{r}_n$ from the last n probes. Each epoch: (i) if $\hat{r}_n > \theta$ (trackable): run AIMD with the safe (α, β) of Prop. 3, converging toward the Bayes kink; (ii) else (untrackable): fall back to the static budget $w_t \equiv \lfloor \bar{a} \rfloor$. The threshold θ is set from the target ϵ via (3) (we reserve γ for the chain's spectral gap in Thm. 2).

Proposition 3 (Sawtooth realization of the kink)**.** *Under stationary capacity a and (A1)–(A3) of the tracking regime, AIMD forms a periodic sawtooth about a with long-run over-admission $\rho = D\alpha/\big((1-\beta)(a + D\alpha)\big) \approx D\alpha/((1-\beta)a)$; choosing $\alpha \le \epsilon(1{-}\beta)a/D$ gives $\rho \le \epsilon$, and $\rho \to 0$ as $D \to 0$.*

Proof sketch. After a backoff the window restarts at $\beta(a + D\alpha)$ and rises by α per clean epoch, overshooting a for D epochs before detection triggers the next backoff at peak $a + D\alpha$; the fraction of violating epochs per cycle gives ρ. (This is the dynamic approach to the static-capacity kink of Thm. 1; D enters both here and in (3), the systems↔theory hinge the paper turns on.) $\square$

Theorem 2 (Uniform near-optimality)**.** *Let capacity be a symmetric two-state process with coherence time τ_c unknown to the guest. The empirical correlation is consistent, $\hat{r}_n \to r(D)$ a.s. (ergodicity), with $\Pr[|\hat{r}_n - r(D)| > t] \le 2e^{-cnt^2}$ for a mixing constant $c > 0$ (Hoeffding for functions of a geometrically mixing chain). Hence the regime-adaptive controller attains the feedback advantage $\Delta(D)$ up to $O(\sqrt{\log n/n})$ (i.e. comes within that error of the age-D optimum)* uniformly in *τ_c, whereas (i) fixed AIMD's excess over-admission stays bounded below by a positive constant as $\tau_c/D \to 0$, and (ii) the fixed static budget forfeits a positive constant of goodput as $\tau_c/D \to \infty$. The regret rate is $O(D\sqrt{\log N/(\gamma N)})$ over horizon N, with γ the chain's spectral gap (App. 0a).*

Proof sketch. Consistency and the concentration bound give a correct regime decision after $n = O(\log(1/\delta)/(r{-}\theta)^2)$ probes w.p. $1 - \delta$; conditioned on the correct branch, Cor. 1 (static branch, where feedback is worthless) and Prop. 3 (AIMD branch) bound the per-epoch gap, and averaging over the $O(1)$ misclassified prefix yields the stated regret. The separations (i)–(ii) are the two endpoints of Cor. 1: at small τ_c AIMD sits at $\rho \approx \frac{1}{4}$ while the optimum is $\rho \to 0$; at large τ_c static forfeits $\Delta \to \frac{1}{2}$. $\square$

a) Scope, honestly.: Three assumptions earn the closed form. (i) *Binary/symmetric per layer:* for any symmetric stationary x_t the value $\frac{1}{2}r(D)$ of (3) is *achievable*—by the memoryless threshold $m_t = \hat{x}_t$, which uses only the freshest sample—and it is the *exact* feedback advantage when x_t is Markov, since then $\hat{x}_t$ is a sufficient statistic for x_t (Thm. 1).

For a general non-Markov symmetric process a history-using controller can do strictly better than $\frac{1}{2}r(D)$—e.g. a random-phase deterministic square wave of period $4D$ has $r(D)=0$ yet its phase, hence x_t exactly, is recoverable from the delayed history, so a causal controller attains $\Delta = \frac{1}{2}$, not 0—so the exact equality is Markov-specific and the general *converse* is the β-mixing bound of App. A, not $r(D)$. Prop. 2 lifts the construction to any integer C once over-admission is scored by *amount* rather than by the $0/1$ indicator (the two coincide at $C=2$, so our testbed is unaffected). (ii) *Stationarity:* $r(D)$ is a lag-D autocorrelation of a stationary process; under slow drift it is replaced by a local estimate, which is exactly what the regime-adaptive controller tracks (§V). (iii) *Fixed age D:* if the freshest sample has a random age A whose value the controller can read, the advantage is the age-averaged $\Delta = \frac{1}{2}\,\mathbb{E}[r(A)]$—the same law under the age law—and the impossibility $\Delta \to 0$ persists whenever $A \gg \tau_c$ almost surely. A matching lower bound in the process's β-mixing coefficient, for general non-binary capacity, is deferred to the appendix. The point stands: **politeness under age-limited sensing is worth exactly what the neighbour's load is predictable across the sensing delay, and a controller should measure that predictability and adapt.**

VI. Realization: Sensing over a Federated Fabric

The controller runs *in-cluster*, behind the network boundary, and reaches the home node over a NATS [13] *leaf* link that needs only one outbound port (Fig. 1): it probes cluster occupancy and actuates admission across that link, submissions flowing on `burst.submit` and results returning on `results.>` by interest propagation, with no inbound cluster access. What matters for the controller is that this view is *aged*—the leaf round-trip and the CUDA cold-start of a freshly admitted pod together set the sensing age D of §IV, which we measure in §VIII. To keep that loop fresh the tensors crossing the link are compressed by random-rotation quantization [14], [15] (fewer bytes, smaller D). The fabric's own design—leaf-federation tradeoffs, JetStream durability, single-port traversal—and the transport and node-level thermal subsystems are a systems contribution developed and characterized in a companion paper [2]; here the fabric is simply the medium that makes capacity sensing possible, and stale.

VII. Implementation

PoliteBurst is a Go control plane (decider, prober, submitter, NATS bridge, back-off modules) plus a Python client and an IPython cell magic [16]. The controller runs *in-cluster* under a minimal namespace-scoped ServiceAccount (Job create/get/delete; pod read), connects to the in-cluster NATS leaf, and renders each `JobDescriptor` into a Kubernetes Job; node-targeting (a `nodeSelector` on the descriptor) pins guest pods to non-reserved GPU types. We verified the full path live on NRP Nautilus [1]: a submit from the home node propagated hub→leaf→controller and launched an A10 GPU worker that ran real CUDA matmul and published its result back through the leaf to the home driver. The system, the controller/worker container images, and the transport/right-sizing packages [14], [17] are available as open-source artifacts (repositories anonymized for review).

VIII. Evaluation

We evaluate the controller on a testbed whose *sensing age is measured, not assumed*: a home node and a workstation-class node with GV100 GPUs, the controller probing real GPU occupancy and admitting real CUDA workers, with a competing tenant driving exogenous capacity. The federated fabric that carries this control plane—and its latency, durability, and scaling characterization—is the companion systems paper's [2]; here we take the fabric as given and measure what the *controller* does over it. Unless noted we report the mean and a t-based 95% confidence interval over independent repeats. The study answers four questions: **(Q1)** the prober detection delay D that governs the bound of Prop. 3; **(Q2)** whether the controller is polite-efficient under a self-imposed budget (the *abundance* regime); **(Q3)** whether it holds under *real, time-varying contention* (the *scarcity* regime Prop. 3 targets)—and where it fails; and **(Q4)** whether the feedback-usefulness law $\Delta(D) = \frac{1}{2}r(D)$ (Thm. 1) holds at the controller level and unifies the two contention regimes as points on one curve.

A. Detection delay and the violation bound (Q1)

D has two measured components. The prober's per-call cost (read-only `nvidia-smi`) is 27–33 ms (p50). The *end-to-end* detection delay —from launching a real GPU load until it is visible as utilization above the 40% floor (10 reps on a GV100, thermally guarded)—is $2.46\,\mathbf{s}$ (mean; p50 2.48, range 2.29–2.61), dominated by CUDA cold-start rather than the poll. The relevant D depends on the question: detecting a *newly admitted* pod's (un)productivity incurs the $\sim 2.5\,$s cold-start term, while probing *already-running* pods costs only the $\sim$ 27 ms poll. In the epoch-based model of §IV one epoch is a single completion-gated admission decision, so the cold-start-dominated age maps to $D \approx 3$ epochs —the value swept in the law validation (Q4)—while the 27 ms poll is sub-epoch for already-running pods. Substituting into Prop. 3, $\rho \approx D\alpha/((1-\beta)\bar{a})$, the $2.46\,$s term makes D a *first-order* contributor to the achievable compliance—not a rounding error—which is precisely the systems↔theory hinge the bound is meant to expose: a guest that admits aggressively (large α) pays for it in over-admission scaled by this measured cold-start delay.

B. Politeness under abundance (Q2)

We burst B jobs under three policies—*naive* (submit up to the hard cap), *static* (fixed submit rate), and *aimd* (the completion-gated admission controller of §V)—against a self-imposed politeness budget C, measuring goodput (tasks/s) and the over-budget fraction ρ (epochs with concurrency $> C$). To isolate the policy from the exogenous scheduling and cold-image-pull variance of the shared cluster—which, reported honestly, produced three pre-registered nulls in situ—we run

TABLE I
END-TO-END GOODPUT VS. POLITENESS (4 REPS, MEAN $\pm$ 95% CI). AIMD HOLDS THE BUDGET ($\rho = 0$) AT STATIC'S POLITENESS BUT 1.7–3.2$\times$ ITS GOODPUT, AND MATCHES NAIVE'S GOODPUT WITHOUT OVER-ADMITTING.

scale	policy	goodput (tasks/s)	over-budget ρ
GPU, C=2	naive	0.0538 ± 0.0003	0.54 ± 0.01
	aimd	0.0539 ± 0.0001	0.00 ± 0.00
	static	0.0313 ± 0.0000	0.00 ± 0.00
CPU, C=8	naive	0.5779 ± 0.0062	0.95 ± 0.00
	aimd	0.2928 ± 0.0029	0.00 ± 0.00
	static	0.0920 ± 0.0002	0.00 ± 0.00

TABLE II
POLITENESS UNDER REAL CONTENTION (C=2; 4 SEEDS, MEAN $\pm$ 95% CI). "KEPT" = COMPETITOR THROUGHPUT RETAINED (1=UNDISTURBED). STRUCTURED (SQUARE): AIMD IS THE POLITE-EFFICIENT KNEE. MEMORYLESS (POISSON): FEEDBACK BREAKS EVEN WITH STATIC—A PRE-REGISTERED NULL.

contention	policy	goodput (tasks/s)	ρ	kept
square	naive	0.101 ± 0.002	0.64 ± 0.01	0.62
	aimd	0.091 ± 0.000	0.15 ± 0.01	0.76
	static	0.064 ± 0.000	0.11 ± 0.01	0.89
poisson	naive	0.096 ± 0.002	0.85 ± 0.06	0.62
	aimd	0.062 ± 0.006	0.30 ± 0.06	0.55
	static	0.064 ± 0.000	0.22 ± 0.04	0.79

on a controlled, variance-free testbed (local-process workers on the workstation node; no scheduler or registry jitter), pre-registered with randomized interleaved trial order and four repeats per cell, at two scales (GPU pool C=2; CPU pool C=8).

Table I and Fig. 3 report the result. **aimd is the polite-efficient frontier point**: it holds the budget exactly ($\rho = 0$ at both scales) while delivering 1.72$\times$ (GPU) and 3.18$\times$ (CPU) the goodput of static at the *same* politeness, and it matches naive's goodput without ever exceeding the budget. naive achieves its goodput only by over-admitting ($\rho = 0.54$ / 0.95); static stays polite but under-utilizes the budget. Both pre-registered hypotheses—H1 (aimd > static goodput at equal ρ) and H2 (naive impolite, $\rho > 0$)—hold with disjoint 95% CIs at both scales.

Scope. This isolates the admission *algorithm* on a controlled node; the full pod-scheduling path (controller $\rightarrow$ node-targeted GPU worker $\rightarrow$ federated result) is verified end-to-end on NRP Nautilus separately (§VII). Here $\bar{a}$ is a fixed self-imposed budget—the *abundance* regime, politeness as pure self-restraint. §VIII-C removes that assumption, driving a_t with a real competing tenant so the Prop. 3 bound is exercised under genuine *scarcity*.

C. Politeness under real contention (Q3)

Abundance tests self-restraint; the harder question is politeness under *scarcity*. We manufacture controlled contention on a two-GPU node (C=2): a *competing tenant* occupies $c(t) \in \{0, 1\}$ GPUs on a fixed profile, so the guest's available capacity $a(t) = C - c(t)$ is exogenous, time-varying, and *measured* by the prober (a per-GPU process census, subject to the detection delay D of Q1)—the regime Prop. 3 targets. Two profiles bracket the space: *square* (a period-2τ wave, modelling slowly-varying or scheduled load) and *poisson* (memoryless on/off arrivals, the adversarial case). Alongside goodput and ρ we measure *harm to the neighbour*: the competitor's completed work relative to its undisturbed baseline. Pre-registered (H1–H3), randomized interleaved order, four seeds per cell.

Table II and Fig. 4 report a sharper picture than abundance did. **Under structured contention (square), aimd is the polite-efficient knee**: 1.41$\times$ static's goodput (0.091 vs 0.064 tasks/s, disjoint CIs) at ρ=0.15—a 4$\times$ lower over-admission than naive's ρ=0.64—while sparing the neighbour (76% of its throughput retained vs naive's 62%). The target ϵ is grounded in the host policy: NRP acts on *persistently* under-utilized requests rather than momentary dips below the $\geq$40% GPU floor, so ϵ is the tolerable share of epochs a pod may sit under-utilized; read that way the ρ=0.15 that aimd holds is a deliberately conservative operating point, inside a $\sim$0.1–0.2 band consistent with that floor at our epoch length. naive's higher goodput (0.101) is bought *entirely* at the neighbour's expense: it over-admits two-thirds of the time and steals 38% of the competitor's throughput (Fig. 5). H1 and H2 hold with disjoint CIs.

Under unpredictable contention (poisson), feedback breaks even—and we report it. aimd's goodput (0.062) is statistically indistinguishable from static's (0.064, overlapping CIs) and it over-admits *more* (ρ=0.30), because capacity now varies on the timescale of the detection delay: by the time the prober sees a competitor arrive, the observation is stale. This is a pre-registered *null* for H1, and it is precisely what Thm. 1 predicts: the two profiles are two coherence times on the $\Delta(D) = \frac{1}{2}r(D)$ curve (Cor. 2), with the slow square ($\tau_c \gg D$) giving feedback room to help and the memoryless poisson ($\tau_c \lesssim D$) driving $r(D) \rightarrow 0$, where no causal controller can beat static—so feedback's break-even is a theorem, not a tuning failure. **(H3)** The measured detection path ($D \approx$ 2–3 s, Q1) also places the Prop. 3 sawtooth ceiling above the observed aimd ρ in both profiles: conservative for a discrete, completion-gated controller, but its prediction—politeness degrades as contention outpaces D—is borne out. Practically, this is exactly the case for the regime-adaptive controller (§V, Thm. 2): track $r(D)$ and fall back to a fixed budget when the neighbour has no structure to exploit. naive is the most impolite in *every* regime (ρ=0.64–0.85).

D. Validating the feedback-usefulness law (Q4)

The contention results above are two points; Thm. 1 predicts the whole curve. To test it we simulate the actual window controllers on the two-state capacity process the theory analyzes—a Markov competitor with per-epoch flip probability p, so the coherence time $\tau_c = -1/\ln(1 -$

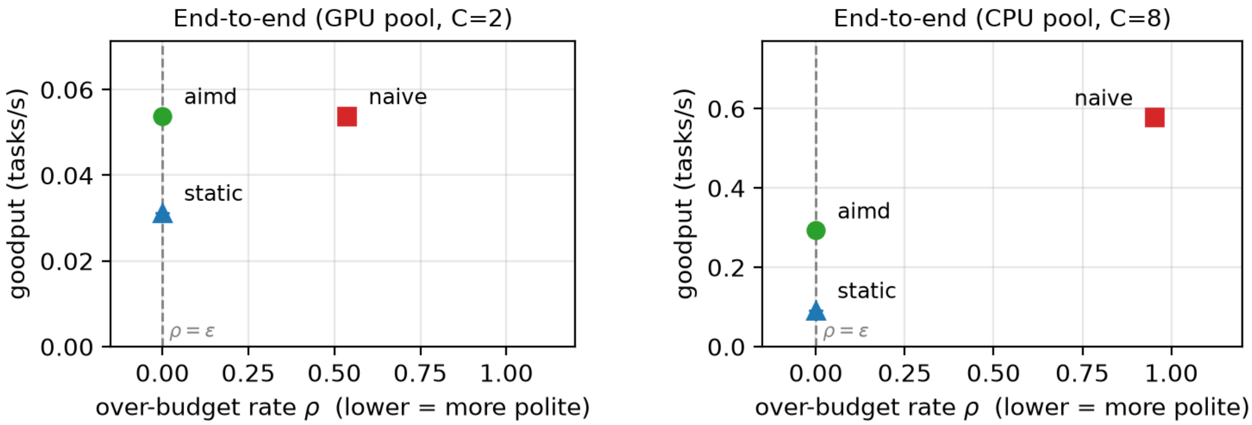


Fig. 3. End-to-end goodput vs. over-budget ρ. *Left:* GPU pool (C=2)—aimd attains naive's goodput at ρ=0 and dominates static. *Right:* CPU pool (C=8)—with surplus capacity the separation widens (aimd delivers 3.18× static's goodput at ρ=0), while naive buys goodput only by flagrant over-admission (ρ=0.95). Dashed line: politeness feasibility bound $\rho = \epsilon$.

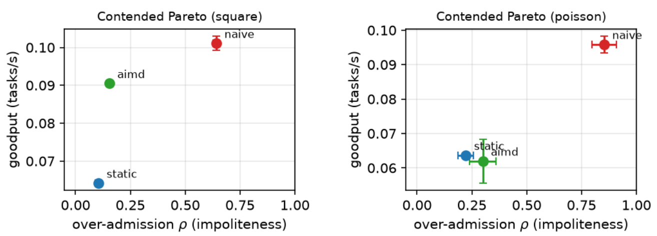


Fig. 4. Contended goodput–politeness frontier (lower-right is better). *Left* (square, structured): aimd is the polite-efficient knee. *Right* (poisson, memoryless): feedback collapses onto static.

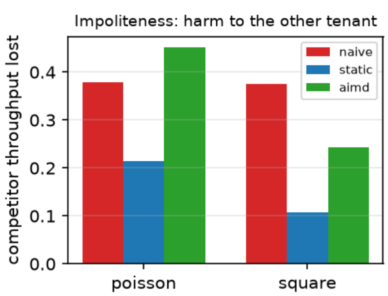


Fig. 5. Harm to the other tenant (throughput lost). naive steals ∼38% in both regimes; aimd spares the neighbour under structured contention (24%), but under memoryless contention its mistimed admissions harm it as much as—here more than—naive (45%): the same D-limited failure as its goodput.

$2p$) sweeps continuously (matching `contention.py --profile markov`)—at the measured sensing age D=3 epochs, $1.2{\times}10^5$ epochs per point. Fig. 6 reports the goodput advantage of each controller over the best static budget at matched over-admission, against the predicted $\Delta(D) = \frac{1}{2}r(D)$.

Three facts confirm the theory and motivate the adaptive controller. **(i)** The delayed-threshold (Bayes-optimal) policy sits *on* the predicted line across three decades of τ_c—

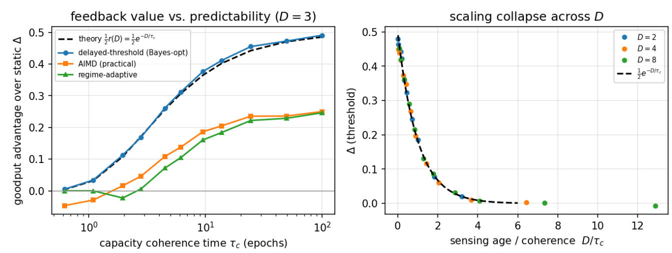


Fig. 6. Controller-level validation of the feedback-usefulness law. *Left:* goodput advantage over static vs. capacity coherence time τ_c at D=3. The Bayes-optimal controller lies on the predicted $\frac{1}{2}r(D)$ (dashed) to within 0.013; practical AIMD falls below static once $\tau_c \lesssim D$; the regime-adaptive controller tracks the envelope. *Right:* sweeping D collapses the optimal advantage onto $\frac{1}{2}e^{-D/\tau_c}$ vs. D/τ_c.

maximum deviation $|\Delta_{\text{opt}} - \frac{1}{2}r(D)| = 0.013$—so the closed form is not an asymptote but the exact frontier. **(ii)** Practical AIMD captures roughly half the available advantage when $\tau_c \gg D$ and, crucially, goes *negative* (worse than static) once $\tau_c \lesssim D$: it keeps probing a signal that has become noise, paying over-admission for nothing—the same failure the live poisson regime shows. **(iii)** The regime-adaptive controller, estimating $r(D)$ online, tracks the upper envelope: it never drops below static and recovers most of AIMD's advantage where feedback is worth having. Finally, sweeping D collapses every curve onto $\frac{1}{2}e^{-D/\tau_c}$ when plotted against D/τ_c (Fig. 6, right), confirming the predicted scaling. The live square/poisson pair (Q3) are the two ends of this axis: $\tau_c \gg D$ (feedback wins) and $\tau_c \lesssim D$ (feedback breaks even), exactly as Cor. 2 states.

a) Live confirmation on GPU hardware.: We ran the regime-adaptive controller on the same 2×GV100 node across the Markov τ_c sweep ($p \in \{0.01, \ldots, 0.4\}$, D=3, two seeds), estimating $r(D)$ online. Two claims transfer from the model. **(A1)** The online estimate lands on the law: $\hat{r} = (1-2\hat{p})^D$ tracks $(1-2p)^D$ with $|\hat{r} - r| = 0.07$ (Fig. 7, left). **(A2)** The switch modulates as predicted, its closed-loop fraction falling

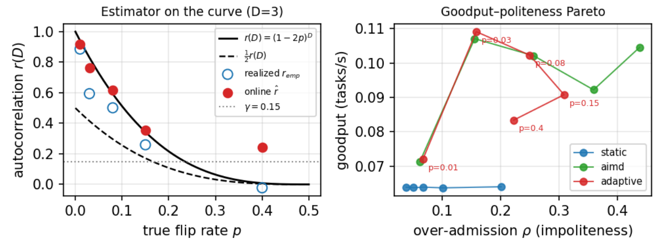


Fig. 7. Live regime-adaptive controller on 2×GV100 (Markov τ_c sweep, D=3, two seeds). *Left:* the online estimate $\hat{r}$ and the realized availability autocorrelation r_{emp} land on $(1-2p)^D$; the closed-loop/static switch fires as $r(D)$ crosses the threshold θ. *Right:* the goodput–over-admission Pareto—adaptive tracks AIMD when feedback is useful and peels off toward static's politeness as p grows, shedding over-admission exactly where $r(D)\to 0$.

$0.98 \to 0.48$ as $r(D)$ drops past the threshold θ. Hardware differs instructively: raw goodput *rewards over-admission*, so live AIMD out-goodputs static in every regime—but at high flip rate that goodput is bought with impoliteness (ρ=0.44 vs 0.04), not tracking. The simulated goodput crossover thus does not appear in *raw* goodput; the law instead surfaces on the goodput–politeness Pareto (Fig. 7, right) as the over-admission the adaptive switch *sheds* versus always-AIMD, growing as feedback loses value ($\rho_{\text{aimd}} - \rho_{\text{adaptive}}$: $+0.01 \to +0.22$ as p : $0.08 \to 0.4$). Adaptive is best-of-both—matching AIMD where the neighbour is trackable and recovering static's politeness where it is not.

E. Summary and threats to validity

The evaluation establishes: (i) the prober detection delay D is a measured, first-order term in the bound—$\approx$ 2.5 s, CUDA-cold-start-dominated (Q1); (ii) under abundance the controller is polite-efficient at both scales, 1.7–3.2× static's goodput at ρ=0 (Q2); (iii) under *structured* contention it is the polite-efficient knee and spares the neighbour, while under *memoryless* contention feedback breaks even with static—a pre-registered null we report rather than suppress (Q3); and (iv) the feedback-usefulness law $\Delta(D) = \frac{1}{2}r(D)$ holds at the controller level, the optimal policy sitting on it to within 0.013 while the regime-adaptive controller never falls below static (Q4), unifying the two live contention regimes as endpoints of one curve; a live GPU run of that controller confirms the online estimate $\hat{r}$ lands on the curve and the closed-loop/static switch modulates as $r(D)$ falls, with the law surfacing on hardware as a *politeness dividend*—the over-admission the switch sheds versus always-AIMD, growing as feedback loses value—rather than a raw-goodput crossover, since raw goodput on real hardware rewards over-admission (Fig. 7). Principal threats: contention is manufactured on a two-GPU node with C=2 and synthetic profiles, not a production multi-tenant trace; the clean goodput-*advantage* tracing of the law (Q4) remains controller-level, as live static is a fixed trickle with no tunable ρ frontier to measure advantage against; and the live adaptive sweep is two seeds per point, a confirmatory anchor rather than the primary evidence. The full system path is nonetheless verified end-to-end (§VII), and naive's impoliteness ($\rho \geq 0.64$, $\geq$38% neighbour harm) is robust across every regime.

a) The law across domains.: Theorem 1 assumes a symmetric two-state *Markov* capacity—is $\Delta(D) = \frac{1}{2}r(D)$ special to GPU bursting, or a property of delayed observation? We tested it on real binary states from other physical domains—acoustic source azimuth (LOCATA) and sperm-whale coda type (Dominica). On each, the value of the best predictor from the delayed *history* (out-of-sample block cross-validation) stays on the single-lag value $\frac{1}{2}r(D)$ and inside a 95% band of 120 Markov surrogates matched on the lag-1 transition (max gaps 0.008 and 0.046; controls: synthetic Markov p=0.44, i.i.d. Δ=0): the single-lag law is exact, not a lower bound. The value of feedback is thus set by the capacity autocorrelation at the sensing lag independent of physical domain. For a capacity predictable beyond one lag (non-Markov) it is only a lower bound, the general converse being the β-mixing bound of App. A.

IX. Related Work

Bursting and function execution. HTCondor [3], funcX/Globus Compute [4], Parsl [5], Ray [6], and Dask [7] provide elastic task execution and provider-driven scaling, but treat the host as owned capacity to fill; none model a fair-use politeness budget or admission under age-limited sensing. They are complementary substrates—a head-to-head dispatch/throughput comparison against them, and the fabric that carries our control plane, is reported in the companion systems paper [2]. **Messaging.** The control plane rides NATS [13] leaf federation rather than a bespoke RPC layer. **Congestion and scavenger control.** Our controller is AIMD in the lineage of additive-increase/multiplicative-decrease congestion avoidance [18], and its *polite* objective—yield to incumbents, use only slack—aligns with low-priority/scavenger transports such as LEDBAT [19] and TCP-Nice [20]. Those protocols are engineered to back off under one-sided loss or delay signals; we differ in two ways. First, the objective is *two-sided* (stay under a pod cap *and* above a utilization floor), which admits an over-admission metric rather than pure loss. Second, and more fundamentally, we do not merely propose a back-off rule: we prove a *limit* on what any such rule can achieve. **Age of information.** That limit is an information-freshness statement—the value of feedback is set by the age D of the capacity estimate relative to the neighbour's coherence time—placing polite bursting in the age-of-information line of work [21] but on the *control* rather than the sampling side: we bound the achievable goodput–politeness frontier as a function of D and give a controller that adapts to the measured regime. To our knowledge, no prior work frames shared-cluster bursting as admission control under age-limited sensing, proves a feedback-usefulness limit for it, or ties an over-admission bound to a measured detection delay.

X. Conclusion

Being a good guest on a shared cluster is a control problem, and how well it can be solved is fixed by how stale the guest's view of the cluster is. PoliteBurst makes that precise: an age-of-information formulation, a feedback-usefulness limit $\Delta(D) = \frac{1}{2}r(D)$ that says closed-loop admission is worth exactly the autocorrelation of capacity at the sensing lag, and a regime-adaptive controller that measures its own predictability and tracks the limit. The limit is validated at the controller level (the optimal policy sits on $\frac{1}{2}r(D)$ to within 0.013; AIMD falls below static precisely when $\tau_c \lesssim D$) and on a workstation testbed under real GPU contention (the full submission path verified end-to-end on NRP Nautilus): under structured contention feedback admission is the polite-efficient knee—$1.41\times$ static's goodput at a quarter of naive's over-admission while sparing the competing tenant a greedy policy harms—and up to $3.2\times$ under a self-imposed budget. The one regime where feedback breaks even, under memoryless contention that outpaces detection, is not a disappointment but the theory's own prediction ($r(D) \to 0$, Cor. 1)—and the regime the adaptive controller detects and sidesteps. Open problems remain: a production multi-tenant trace (ours is manufactured contention at $C{=}2$), extending the exact limit beyond the binary layer to general capacity processes via the β-mixing lower bound, and predictive controllers that estimate $r(D)$ rather than only its regime.

APPENDIX

Corollary 1 showed feedback's value vanishes with the sensing age D for the two-state model. We now show the same for *any* stationary capacity process, with the decay rate governed by the process's β-mixing (absolute regularity) coefficient—so the exact binary law of Thm. 1 is one tight instance of a general information-theoretic limit.

a) Setup.: Let $\{a_t\}$ be stationary on $\{0, \ldots, C\}$ with $\mathcal{F}_{\le s} = \sigma(a_u : u \le s)$ and $\mathcal{F}_{\ge s} = \sigma(a_u : u \ge s)$. Its β-*mixing coefficient* at lag D is

$$\beta(D) = \mathbb{E}\left[\sup_{B \in \mathcal{F}_{\ge t}} \left| \mathbb{P}(B \mid \mathcal{F}_{\le t-D}) - \mathbb{P}(B) \right| \right], \quad (5)$$

with $\beta(D) \to 0$ iff the process is absolutely regular; geometrically mixing processes satisfy $\beta(D) \le c_0 \varrho^D$ for some $\varrho \in (0, 1)$. (The β-mixing coefficient always carries its lag argument $\beta(D)$; it is unrelated to the AIMD back-off factor β of the controller.) Fix any bounded per-epoch objective $R(w, a)$—e.g. the politeness Lagrangian $R_\lambda(w, a) = \min(w, a) - \lambda(w - a)^+$—with oscillation $\Omega \triangleq \max_{w,a} R - \min_{w,a} R$. A causal age-$D$ controller sets w_t measurable w.r.t. $\mathcal{G} \triangleq \mathcal{F}_{\le t-D}$; a static controller uses a $\mathcal{G}$-independent w. Write $V^\star(D)$, V_{stat} for their optimal per-epoch values and $\Delta(D) = V^\star(D) - V_{\text{stat}}$.

Lemma 2 (Lipschitz value function)**.** $\psi(\mu) \triangleq \max_w \sum_a R(w, a)\,\mu(a)$ *is convex on the simplex and* Ω-*Lipschitz in total variation:* $|\psi(\mu) - \psi(\mu')| \le \Omega\,\|\mu - \mu'\|_{\text{TV}}$. *(Convexity is the classical source of the nonnegativity of information value [10]; the Lipschitz-in-TV bound is the payoffs–beliefs duality of [11]. Theorem 3 below is thus the standard value-of-information bound specialized to the mixing coefficient.)*

Proof. ψ is a pointwise maximum of linear functionals, hence convex. Let w_μ attain $\psi(\mu)$. Then $\psi(\mu) - \psi(\mu') \le \sum_a R(w_\mu, a)\,(\mu - \mu')(a) \le \text{osc}_a R(w_\mu, \cdot)\,\|\mu - \mu'\|_{\text{TV}} \le \Omega\|\mu - \mu'\|_{\text{TV}}$, using $\sum_a f(a)\nu(a) \le \text{osc}(f)\|\nu\|_{\text{TV}}$ for zero-sum ν; the reverse bound is symmetric. □

Theorem 3 (β-mixing impossibility)**.** *For any stationary capacity process and any bounded objective,*

$$0 \;\le\; \Delta(D) \;\le\; \Omega\,\beta(D). \quad (6)$$

Hence no causal age-D controller outperforms static as $D \to \infty$ whenever the capacity is absolutely regular; if $\beta(D) \le c_0\varrho^D$ the advantage decays geometrically, $\Delta(D) \le \Omega c_0 \varrho^D$.

Proof. Let $\mu_{\mathcal{G}} = \mathbb{P}(a_t \in \cdot \mid \mathcal{G})$; the tower property gives $\mathbb{E}[\mu_{\mathcal{G}}] = \bar{\mu} \triangleq \mathbb{P}(a_t \in \cdot)$. Optimizing the $\mathcal{G}$-measurable window pointwise, $V^\star(D) = \mathbb{E}[\psi(\mu_{\mathcal{G}})]$ and $V_{\text{stat}} = \psi(\bar{\mu})$, so convexity of ψ and Jensen give $\Delta(D) = \mathbb{E}[\psi(\mu_{\mathcal{G}})] - \psi(\bar{\mu}) \ge 0$ (freshness never hurts). For the upper bound, Lemma 2 gives $\Delta(D) \le \Omega\,\mathbb{E}\|\mu_{\mathcal{G}} - \bar{\mu}\|_{\text{TV}} = \Omega\,\mathbb{E}[\sup_B |\mathbb{P}(a_t{\in}B \mid \mathcal{G}) - \mathbb{P}(a_t{\in}B)|] \le \Omega\beta(D)$, the last step because $\sigma(a_t) \subseteq \mathcal{F}_{\ge t}$ so (5) takes the supremum over a larger field. □

Corollary 3 (Binary tightness)**.** *For the symmetric two-state model, $\beta(D) = \frac{1}{2}|1 - 2q(D)| = \frac{1}{2}r(D)$ (a standard mixing identity [12], [8]), so Thm. 3 reproduces the exact advantage of Thm. 1 up to the objective scale Ω.*

Proof. The Markov property reduces $\mathcal{F}_{\le t-D}$ to a_{t-D}; the conditional-to-marginal TV distance is $|\frac{1}{2} - q(D)|$ from either state, so $\beta(D) = \frac{1}{2} - q(D) = \frac{1}{2}r(D)$. □

We sketch the uniform near-optimality asserted in Thm. 2. The controller estimates $\hat{p}_n = \frac{1}{n}\sum_{i=1}^n \mathbf{1}[\hat{a}_i \ne \hat{a}_{i-1}]$, forms $\hat{r}_n = (1 - 2\hat{p}_n)^D$, and runs the closed-loop branch iff $\hat{r}_n > \theta$. The flip indicators are a bounded functional of the reversible two-state chain (absolute spectral gap $\gamma = 1 - |1 - 2p| = 2\min(p, 1 - p)$), so the spectral-gap Hoeffding inequality for reversible chains [22], [23] gives $\Pr(|\hat{p}_n - p| \ge t) \le 2e^{-c_0\gamma n t^2}$ for a universal $c_0 > 0$. As $p \mapsto (1 - 2p)^D$ is $2D$-Lipschitz on $[0, \frac{1}{2}]$, $|\hat{r}_n - r(D)| \le 2D|\hat{p}_n - p|$; hence with a threshold margin $|r(D) - \theta| \ge \Delta\theta$ the branch is wrong with probability $\le 2\exp(-c_0\gamma n(\Delta\theta)^2/4D^2)$.

Theorem 4 (Uniform regret)**.** *Set θ at the branch* indifference *level, where the two branches' values are L-Lipschitz in $r(D)$ and coincide. Then the regime-adaptive controller's time-averaged value over horizon N obeys $\sup_{\tau_c} \text{Regret}_N = O(D/\sqrt{\gamma N})$, and $\text{Regret}_N = O(1/N)$ for any fixed process with $r(D) \ne \theta$.*

Proof sketch. A wrong branch costs at most $L\Delta\theta$; averaging the wrong-branch bound gives $\text{Regret}_N \le L\min(\Delta\theta,\, 8D^2/(c_0\gamma N\Delta\theta))$, which is $O(1/N)$ for fixed $\Delta\theta$ and, maximized over τ_c at $\Delta\theta^\star = \sqrt{8D^2/(c_0\gamma N)}$, yields $O(D/\sqrt{\gamma N})$. □

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